



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-27
Course Handout Part II

Date: 3rd Aug 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number: CE F331

Course Title: In-situ Testing Methods in Geotechnical Engineering

Instructor-in-charge: Dr. Sayantan Chakraborty

Course description: In-situ Testing Methods in Geotechnical Engineering is an introductory course to various in-situ test methods pertaining to geotechnical engineering and analysis of in-situ test results, including drilling and sampling, standard penetration test (SPT), cone-penetration test (CPT), piezocone penetration test (CPTu), vane shear test (VST), non-destructive tests (NDT) using body waves and surface waves - seismic reflection test, seismic refraction test, and spectral analysis of surface waves (SASW) test. Structure and components of geotechnical reports and data presentation techniques using borehole logs will be discussed. The contents of this course will enable students to utilize different in-situ test results for soil property interpretation, direct and indirect applications for the design of geotechnical structures, and liquefaction assessment. This course will be helpful to students when they start working as practicing engineers or for further graduate-level advanced courses in geotechnical engineering.

Scope and Objective: Introduction to in-situ testing, drilling, and sampling, advantages and disadvantages of in-situ testing, and scope of using the test results for direct and indirect design of geotechnical structures.

- Learn about SPT, test procedure, SPT corrections, and frequently used correlations to estimate different geomaterial properties.
- Become familiar with some of the frequently used methods for the design of foundations using SPT data.
- Learn about CPT, test procedure, and frequently used correlations to estimate different geomaterial properties.
- Design of foundations based on CPT results.
- Assess liquefaction potential for a site based on SPT and CPT results.
- Estimate seismic wave velocities and interpret subsurface profiles based on seismic reflection and refraction tests, and SASW test.
- Structure and components of geotechnical reports and data presentation techniques using borehole logs based on exploration findings will be discussed.

Course Learning Outcome:

- Remember different in-situ testing methods related to geotechnical engineering.



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- Understanding of different widely used in-situ testing procedures and principles pertaining to geotechnical engineering.
- Knowledge of appropriate correlation equations available to interpret different geomaterial properties using in-situ test results.
- Apply different direct design methods for designing foundations.
- Analyze and compare different in-situ test results for tests conducted at the same site and decide suitability of different test methods.
- Evaluate the suitability of different test methods based on site details and purpose of testing, select suitable method for foundation design using in-situ test results and assess potential for liquefaction.

Textbooks and references:

TB1. In-situ Testing Methods in Geotechnical Engineering by Alan Lutennegar

RB1. In-situ Testing in Geomechanics: The Main Tests by Fernando Schnaid

RB 2. Geotechnical Earthquake Engineering by Steven Kramer

RB 3. Manual on Subsurface Investigation (NCHRP W 258)

RB 4. Guide to Cone Penetration Testing by P.K. Robertson and K. Cabal

RB 5. Relevant IS codes

Method of conduct of the course: Lectures using PowerPoint slides

Course Plan: weekly schedule for 14 weeks with 3 lectures per week

Module No.	Topic and Lecture Session Number*	Reference	Learning outcomes
1	Introduction to in-situ testing, drilling, and sampling: L1-L3	TB – Ch 1, RB1 – Ch 1, Lecture PPT	Learn about the role of in-situ tests in site investigations, advantages and limitations, applications of in-situ tests in estimating geomaterial properties, and scope of using the test results for direct and indirect designs. Get familiar with the widely used drilling and soil sampling techniques.
2	Standard penetration test (SPT): L4-L10	TB – Ch 2, RB1 – Ch 2, Lecture PPT	Learn about SPT mechanism, test procedure, condition for refusal, factors affecting SPT N value, corrections of SPT field N value, and SPT correlations.
3	Foundation design using SPT data: L11-L13	TB – Ch 2, RB1 – Ch 2, Lecture PPT	Learn about shallow and deep foundation design methods using SPT data.
4	Cone penetration test (CPT): L14-L24	TB – Ch 4, RB1 – Ch 3, Lecture PPT	Learn about cone penetration test method, its working principle, interpretation of CPT and CPTu data, advantages and disadvantages of CPT, and CPT correlations. Get familiar with seismic cone penetration test



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			(SCPT) procedure and estimation of shear wave velocity profile based on SCPT data.
5	Foundation design using CPT: L25-L26	TB – Ch 4, RB1 – Ch 3, Lecture PPT	Learn about different methods used for foundation design using CPT data - direct method to estimate bearing capacity and settlement.
6	Liquefaction assessment using SPT and CPT data: L27-L28	TB Ch – 2 and 4, RB1 Ch – 2 and 3, Lecture PPT	Assessment of liquefaction potential of a site using SPT and CPT data.
7	NDT using seismic waves - seismic reflection and refraction tests and SASW test: L29-L39	RB2 – Ch 6, Lecture PPT	Learn about seismic waves, estimation of moduli using seismic wave velocities, reflection and refraction of seismic waves, reflection coefficient, Snell's law, first arrival time, reflection test, limitations of reflection test, and seismic refraction test. Learn about Rayleigh wave properties, dispersive nature of Rayleigh waves, steady state vibration test, SASW test methodology, and interpretation of SASW test results, distance criteria and useable wavelength for interpretation, masking, global dispersion curve, and inversion analysis
12	Project work and presentation: L40-41	Real in-situ test data will be provided at the beginning of the semester	Characterization of soils based on bore log data, SPT, and CPTu results, estimation of different geomaterial properties using SPT and CPT-based correlations, comparison of results, liquefaction assessment, and direct design using in-situ test data.

Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quiz	30%	30 mins	TBA	Best 5 out of 7, Open book
Project	30%	Throughout the semester take home assignments will be	TBA	Take home assignments, report submission and in-class presentation.



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		given, along with interim evaluation, and end-semester report submission and presentation.		
Comprehensive	40%	180 min	TBA	Open book

Criterion for NC: Missing any evaluation component completely, appearing in the quizzes or examinations for the sake of attendance, and securing less than 10% of total marks.

Chamber consultation hours: Saturdays 10 – 11 am.

Makeup Policy: No make-up exams will be conducted for quizzes or project presentations. Make-up request for Comprehensive examination must be notified before the examination and decision will be made as per institute norms. Make-up requests made on medical grounds may not be approved unless recommended by CMO BITS Pilani.

Instructor-in-charge

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